

Predictive Modeling and Risk Scoring for Bank Customer Churn

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Abstract

One of the biggest problems retail banks face is when customers leave, which can really hurt their business in the long run. This is because it affects how much money they can make from each customer over time, and also how much revenue they can generate. The old way of dealing with this issue is to wait until customers have already decided to leave, which doesn't give the bank much time to do anything about it. This approach also makes it more expensive for the bank to find new customers. So, there's a real need for smart systems that can predict which customers are likely to leave before they actually do. This would allow the bank to step in and try to keep them before it's too late. By doing this, banks can reduce the number of customers they lose and also save money on finding new ones. It's all about being proactive and using data to understand what customers might do, rather than just reacting after they've already made up their minds.

This study looks at how to use machine learning to predict when customers might stop doing business with a bank. It uses a large set of data from 10,000 banking customers in France, Germany, and Spain. The researchers first cleaned up the data and then found ways to make it more useful by looking at how customers interact with the bank, their financial habits, and how they use different bank products. They tried out five different machine learning models to see which one worked best at predicting customer churn. These models were tested using several measures, including how accurate they were, how precise they were, how well they recalled important information, their overall score, and how well they could distinguish between customers who were likely to churn and those who were not.

Among all evaluated models, the Gradient Boosting classifier demonstrated the strongest overall predictive capability by achieving a Test ROC-AUC of 0.8652 with an overall classification accuracy of 86.2%. Furthermore, SHAP (SHapley Additive exPlanations) analysis was employed to improve model transparency by identifying the most influential factors contributing to customer churn. The findings reveal that customer age, number of

banking products, active membership status, and geographical location are the primary determinants of churn risk.

The new approach helps banks move away from just reacting to customers leaving and instead use data to make informed decisions. By spotting customers who are at risk of leaving early on, banks can create targeted campaigns to keep them, make the most of their marketing budget, build stronger relationships with customers, and increase profits in the long run - all while being open and following the rules. This way, banks can focus on keeping customers happy and loyal, rather than just trying to win them back when it's too late.

1. Introduction

The banking world has changed a lot in the last ten years, with lots of new technology and competition between banks. Now, customers have many options to choose from, with good interest rates, easy online services, and financial products that fit their needs. Because of this, it's much harder for banks to keep their existing customers than it used to be. Studies have shown that it costs a lot more to get a new customer than to keep an old one, so banks are really focusing on keeping the customers they already have. This is a big deal for banks that work with regular people, and they're trying to figure out the best ways to do it.

Customer churn refers to the phenomenon in which customers discontinue their relationship with a financial institution by closing accounts or moving their financial activities to competing banks. High churn rates reduce customer lifetime value, decrease revenue generated through deposits and financial products, and negatively affect overall business performance. Consequently, banks require intelligent systems capable of identifying customers who are likely to churn before they actually terminate their relationship.

Traditional churn management systems primarily rely on historical reports and manual analysis, making them reactive rather than predictive. These methods often fail to capture the complex relationships among customer demographics, transaction behavior, product utilization, and account activity. Recent developments in machine learning have enabled organizations to analyze large volumes of customer data and identify hidden behavioral patterns that cannot be recognized using conventional statistical techniques.

This study suggests a new way to use machine learning that helps banks figure out which customers are likely to stop doing business with them. It combines special techniques to make the most of the data, uses several methods to make predictions, and includes a way to

explain the results clearly. Instead of just saying whether a customer is leaving or staying, this approach gives banks a sense of how likely it is that each customer will leave, so they can focus on the ones that are most at risk. This can help banks keep more customers and provide better service to the ones who need it most. By using this framework, banks can make more informed decisions and take targeted actions to reduce customer churn.

This project's main achievement is creating a churn prediction system that's accurate, easy to understand, and focused on business needs. It helps companies keep their customers by predicting who's likely to leave. The tests show that our proposed Gradient Boosting model works well and is reliable, while also being simple enough to use in real-world financial institutions. This means banks and other financial companies can use it to make informed decisions and take action to retain their customers.

2. Related Work (Literature Review)

Customer churn prediction has become an important research area in banking, telecommunications, insurance, and subscription-based industries due to its direct impact on customer retention and organizational profitability. Early studies primarily relied on statistical methods such as Logistic Regression because of their simplicity and interpretability. Although these approaches provided valuable insights into customer behavior, they were often unable to model the complex nonlinear relationships present in modern customer datasets.

As machine learning keeps getting better, researchers started using things like Decision Trees, Random Forests, and Support Vector Machines to make predictions about customer churn more accurate. They also used Gradient Boosting and Extreme Gradient Boosting, which is also known as XGBoost. These ensemble learning methods work really well because they combine lots of decision trees to understand complicated patterns in how customers behave, and they also reduce mistakes in predictions. By using multiple trees, they can capture more details and make better guesses about what customers will do. This helps businesses get a better idea of which customers might leave, so they can try to keep them.

New research is highlighting how important it is to understand artificial intelligence when it comes to making financial decisions. These days, companies have to explain how their automated systems make predictions that affect customers. One way to do this is by using something called SHAP, which helps figure out how each piece of information contributes to

a specific prediction made by a machine learning model. This is a big deal because it helps us trust the decisions made by machines.

Despite these advances, many existing studies focus primarily on improving prediction accuracy while providing limited emphasis on practical business interpretation and customer-specific risk scoring. This research addresses that gap by combining advanced feature engineering, multiple machine learning models, and SHAP-based explainability into a unified framework that supports proactive customer retention and informed decision-making within retail banking environments.

3. Problem Statement and Objectives

Problem Statement

Banks have a lot of information about their customers from things like transactions, account activity, and demographic records. But, a lot of the systems they use to manage customer relationships are still looking at what's happened in the past, rather than trying to predict what might happen next. This means that banks often don't realize a customer is leaving until it's too late. By the time they notice, the customer has already made up their mind to go elsewhere. This approach is not very effective, and it ends up costing banks more money to try to win customers back. If banks could use smarter systems to anticipate when a customer might leave, they could try to keep them earlier on, and save themselves a lot of trouble and expense.

Many old ways of predicting customer behavior don't work well because they can't handle the complicated relationships between how customers act, their financial situation, and how they use products. Also, super accurate machine learning models that are like black boxes often don't give clear explanations, which makes it hard for business leaders and regulators to understand why certain decisions are made. So, we really need a prediction system that is accurate, easy to understand, and practical, which can help companies keep their customers by being proactive. This system should be able to show why it's making certain predictions, so businesses can use that information to make good decisions and keep their customers happy.

Research Objectives

The primary objectives of this research are:

- To develop an accurate machine learning model capable of predicting customer churn in retail banking.

To create personalized scores that show how likely a customer is to stop doing business with a company, so the company can take action to keep them.

- To compare the performance of multiple machine learning algorithms using standard evaluation metrics.
- To identify the most influential factors contributing to customer churn through SHAP explainability analysis.
- To provide practical business recommendations that help financial institutions improve customer retention and optimize marketing resources.

4. Dataset Description

This research uses a big dataset with information from 10,000 bank customers in France, Germany, and Spain. Each customer's record has lots of details like who they are, what kind of account they have, how much money they have, and how much they use the bank's services. The main goal of looking at all this data is to figure out which customers are likely to stop doing business with the bank.

The information we have about our customers is really important and it includes things like how old they are, if they're a man or a woman, where they live, how much money they have in their account, how much we think they make in a year, how long they've been with us, their credit score, how many of our banking products they use, if they're still an active member, and if they have a credit card. We're trying to figure out if a customer is still with us or not, so we have a special variable called "Exited" that tells us this - if it's 0, that means the customer is still our customer, but if it's 1, that means they've left us.

To start with, some information wasn't useful for figuring out if a customer would stop doing business with us. This included things like the customer's ID, last name, and the year, which was always the same. We took these out because they didn't help us understand why customers might leave. Then, we used the remaining information to look deeper into the data, make new features that would be helpful, and build a model that could predict customer churn.

This dataset is really useful for creating models that can predict which customers are likely to leave the bank. The reason is that it has a good mix of information about customers, including who they are, how they behave, and their financial situation. This mix of numbers and categories allows us to use a wide range of analysis tools, from basic statistics to more complex machine learning methods.

5. Exploratory Data Analysis (EDA)

To get a better sense of our customers, we did some exploratory data analysis. This helped us understand how our customers are spread out, find important patterns in how they behave, and see how different factors affect whether they stick with us or leave. We learned some really useful things from this that helped us decide which features to focus on and which models to use.

When we looked at the numbers, we found that almost 80% of customers, to be exact 7,963, stayed with the bank, while about 20%, or 2,037 customers, decided to leave. This kind of situation is pretty common when trying to predict which customers will stop doing business with a company - there are usually a lot more customers who stay than those who leave. So, when we evaluated how well our model worked, we didn't just look at how often it was right, but also at other important measures like precision, recall, F1-score, and ROC-AUC, to make sure we got a fair picture of how it was performing.

When it comes to figuring out why customers leave, one thing that really stands out is the customer's age. It seems that people between 45 and 60 years old are more likely to stop doing business with the bank compared to younger customers. On the other hand, customers who are between 18 and 35 years old tend to have a stronger connection with the bank, which means they're more likely to stick around. This suggests that younger customers are more loyal to the bank, while older customers are more likely to take their business elsewhere.

Analysis of product ownership revealed an interesting behavioral pattern. Customers maintaining exactly two banking products demonstrated the highest retention rates.

Conversely, customers with only one product or more than two products exhibited significantly greater churn probability. This observation suggests that moderate product engagement strengthens customer loyalty, whereas either limited engagement or excessive product complexity may encourage customers to leave.

Geographical analysis indicated substantial variation across different countries. Customers located in Germany experienced noticeably higher churn rates than customers in France and Spain, suggesting possible regional differences in competition, customer satisfaction, or banking policies.

Getting customers involved is really important. When people use banking services regularly, they're less likely to stop using them. This shows that interacting with banking services often can help keep customers long-term. These findings helped us create new features to make our machine learning models better at predicting what customers will do.

6. Analytical Methodology

A structured machine learning methodology was adopted to develop an accurate and interpretable customer churn prediction system. The overall workflow consisted of data preprocessing, feature engineering, model training, model evaluation, and explainability analysis.

Initially, the raw dataset underwent preprocessing to eliminate irrelevant variables and prepare the data for machine learning algorithms. Categorical variables were transformed into numerical representations using one-hot encoding, while numerical variables were standardized using the StandardScaler technique to ensure comparable feature scales. These preprocessing steps improved model stability and prevented bias caused by differences in variable magnitudes.

To improve our ability to make predictions, we created some new features that are specific to our area of work. We did this by combining some of the existing information we have about our customers. These new features help us understand some of the financial connections and patterns of how our customers interact with us that we couldn't see just by looking at the original data.

We tried out five different machine learning methods to see which one worked best. These methods were Logistic Regression, Decision Tree, Random Forest, Gradient Boosting, and XGBoost. We used the same information to train and test each method, so we could compare them fairly and see which one made the best predictions.

To better understand how the best model worked, a special analysis called SHAP was used. This analysis helped figure out how much each piece of information contributed to predicting whether a customer would stop using the service. By doing this, the company could be more

open and honest about how they made their predictions, which made stakeholders more confident and helped the company follow the rules by clearly explaining why a customer got a certain score for being at risk of leaving.

6.1 Data Preprocessing

Getting good results from machine learning models depends a lot on how well the data is prepared. To make sure the predictions are accurate and reliable, several steps were taken to clean up and organize the data before training the models. This helps to improve the overall quality of the models and makes them more trustworthy.

To start with, some information like customer IDs, surnames, and the year weren't used because they don't really help predict if a customer will stop using a service. Keeping this kind of data wouldn't make the prediction model better, it would just make it more complicated.

Categorical variables including Geography and Gender were transformed into numerical values using one-hot encoding. This conversion enables machine learning algorithms to process categorical information without introducing artificial ordering between categories.

To make the numbers work better, we used a special technique called StandardScaler on things like Credit Score, Balance, Age, Estimated Salary, and how long someone's been with us. This helps make all the numbers similar, so they're all around zero and have the same amount of variation. It's like getting everyone on the same page, so the model can learn more efficiently and not get thrown off by really big numbers. This way, all the variables get an equal say in how the model is trained, rather than some of them overpowering the others.

The system got rid of any messy data and made sure everything was consistent and ready for machines to learn from, which is perfect for figuring out when customers might stop doing business with us.

6.2 Feature Engineering

To improve how well machine learning models can predict things, some extra work was done to create new variables that better show how customers behave with their money and how they interact with things. Instead of just using the original information, new features were made to catch important connections between different customer characteristics. This helps the models understand customers better.

The Balance-to-Salary Ratio was introduced to measure the proportion of customer account balance relative to annual income. Customers maintaining unusually high balances compared with their salaries may exhibit different financial behaviors and greater sensitivity to competitive banking services.

To figure out how quickly customers take on new banking products, we looked at the number of products they had and how long they'd been with the bank. We called this the Product Density feature. It's basically a way to measure how fast customers start using more banking products over time.

The company came up with a new feature called Active Product Interaction. This was done by looking at how active the customers were and how many products they owned. What they found out was that owning a product only helps keep customers around if they actually use it.

To better understand customer behavior, a new feature was created by combining two important factors: the customer's age and how long they've been with the company. This combination, called the Age-Tenure Interaction feature, helps the model recognize patterns in how customers behave over time, especially as they move through different stages of life. By doing so, it can identify more complex reasons why customers might stop doing business with the company.

These engineered variables significantly improve the model's ability to capture nonlinear customer behavior that may not be represented through the original features alone.

6.3 Train-Test Strategy

To get a fair idea of how well the model works, the data was split into two parts: one for training and one for testing. This was done using a simple 80:20 rule, where 80% of the data is used for training and 20% for testing. So, around 8,000 customer records were used to train the model, and about 2,000 records were kept separate to see how well the model performs on its own.

A stratified sampling strategy was adopted to preserve the original churn distribution in both subsets. Maintaining similar proportions of churned and retained customers prevents sampling bias and allows evaluation metrics to accurately represent real-world performance.

When we're trying to figure out how customers behave, we use something called a training dataset to learn their patterns and make our model better. But we keep another set of data,

called the testing dataset, separate and don't look at it while we're training. This way, when we test our model, we can see how well it really works with new customer information, rather than just repeating what it's already seen. It's like trying to predict what a new customer will do, and we want to make sure our model is good at that, not just good at remembering what old customers did.

Using a stratified train-test strategy improves the credibility, reliability, and reproducibility of the experimental results presented in this research.

7. Model Development

To figure out which predictive model works best for customer churn, we tried out five different supervised machine learning algorithms and compared their results. We chose these algorithms because they start with simple models and move on to more complex ones that can handle complicated relationships in customer data.

The first model we used was Logistic Regression, which is a simple and easy-to-understand way to classify things into two groups. We chose it as our baseline because it's widely used and easy to work with. However, it has a big limitation: it can't handle complex customer behavior that doesn't follow a straight line.

A special kind of computer program, called a Decision Tree classifier, was created to understand how customers make decisions and how their characteristics affect those decisions. This program is really good at showing how different factors are connected, but it can be a problem if it's used on its own because it can get too complicated and start to make mistakes.

To make predictions more stable, a Random Forest model was used. This model works by combining lots of decision trees, which helps to reduce errors and make the predictions more accurate. It's like having multiple people make a decision, and then taking the average of all their answers. This approach is called bootstrap aggregation, or bagging for short. By doing this, the model can make better predictions than if it just used one decision tree.

The researchers also looked at Gradient Boosting, a way of combining multiple models to make predictions. It works by creating a series of decision trees, with each tree trying to fix the mistakes made by the previous one. This process of learning from mistakes helps the model pick up on small patterns in how customers behave, which can really improve its ability to make accurate predictions.

To finish, a special kind of boosting algorithm called Extreme Gradient Boosting, or XGBoost for short, was used. This algorithm is really good at handling structured datasets and does well in machine learning competitions. It's also very efficient and uses techniques to prevent overfitting, which means it can make good predictions without getting too complicated. XGBoost is a popular choice among machine learning experts because it's powerful and fast.

All models were trained using the same preprocessed dataset and identical train-test split to ensure a fair comparison of predictive performance. Their effectiveness was assessed using Accuracy, Precision, Recall, F1-Score, and ROC-AUC.

8. Model Performance Evaluation

Performance Comparison

Classifier	Accuracy	Precision	Recall	F1-Score	ROC-AUC
Logistic Regression	70.95%	38.27%	69.78%	49.43%	0.7759
Decision Tree	75.80%	44.52%	76.90%	56.40%	0.8261
Random Forest	85.10%	65.18%	57.49%	61.10%	0.8617
Gradient Boosting	86.20%	74.35%	49.88%	59.17%	0.8652
XGBoost	86.90%	77.78%	50.12%	60.78%	0.8621

8.1 Results and Discussion

The tests show that using multiple learning methods together works better than traditional ways of teaching machines to predict when customers will stop doing business with a company. One method, called Logistic Regression, didn't do very well because it assumes that customer characteristics and the likelihood of them leaving are connected in a straightforward way. Even though this method was good at finding customers who were likely to leave, it made a lot of mistakes by falsely identifying customers who weren't going to leave, which makes it less useful for campaigns aimed at keeping customers.

The Decision Tree classifier improved predictive performance by learning nonlinear decision boundaries. However, its tendency to overfit reduced its ability to generalize effectively to unseen customer data.

Both Random Forest and Gradient Boosting achieved considerably higher predictive performance by combining multiple decision trees into ensemble models. Random Forest produced balanced classification results and demonstrated strong robustness against overfitting. Gradient Boosting further enhanced prediction quality by sequentially correcting previous classification errors, resulting in improved probability estimation and better discrimination between churned and retained customers.

Although XGBoost obtained a slightly higher classification accuracy, the Gradient Boosting model achieved the highest ROC-AUC score of 0.8652, indicating superior capability in distinguishing high-risk customers from low-risk customers across different classification thresholds. Since customer churn prediction is fundamentally a probability estimation problem, ROC-AUC provides a more reliable measure of model effectiveness than accuracy alone.

So, the team chose Gradient Boosting for their prediction model because it works really well for figuring out customer risk scores. It's good at making predictions, estimating probabilities, and staying stable, which makes it practical to use in real-world applications. This means it can help businesses make better decisions about their customers.

9. Model Explainability using SHAP

Financial institutions need to be able to trust the decisions made by automated systems, and that means they need to understand how those decisions are made. So, to make things more transparent, a technique called SHAP analysis was used on a type of machine learning model called Gradient Boosting. This helps break down the decision-making process into smaller parts, making it easier to see what's going on behind the scenes.

SHAP values help us figure out how much each feature affects a prediction for a specific customer. This means that both the people analyzing the data and the business managers can understand why a customer got a certain score for being at risk of stopping their service. Being able to understand the reasons behind these predictions makes people trust the machine learning models more and also helps with following rules that require artificial intelligence to be explainable.

The study found that how old a customer is has a big impact on whether they stop doing business with a company. It turns out that customers who are between 45 and 60 years old are more likely to leave compared to younger customers. This was shown by a special analysis called SHAP, which helps figure out what factors are most important in predicting customer behavior.

It seems that the number of products a customer has can really affect how likely they are to stop doing business with a company. When customers have exactly two banking products, they are less likely to leave. But if they only have one product or more than two, they are more likely to go elsewhere. This is an important thing to consider when trying to keep customers happy and prevent them from leaving.

Regular interactions with customers play a big role in keeping them loyal. When customers don't use their accounts much, they're more likely to stop being customers altogether. This shows that banks need to keep in touch with their customers to build strong, long-lasting relationships.

Where someone lives can also affect whether they switch banks. For example, people living in Germany are more likely to switch than those in France and Spain. Also, customers with a lot of money in their accounts compared to how much they earn in a year are more likely to be tempted by other banks' offers and leave their current bank. This is because they have more to lose if they're not getting the best deal, so they're always on the lookout for something better.

These results show that using SHAP analysis does two important things: it makes models more transparent and it gives us useful ideas for keeping customers that are tailored to each individual.

10. Strategic Recommendations

To help retail banks keep their customers and stop them from going to other banks, some useful ideas can be tried. These ideas come from looking at the results of predictive modeling and analyzing which features are most important using SHAP.

Banks need to focus on the customers that are likely to leave, as shown by their predictive models. They should try to keep these customers by creating special campaigns just for them, before it's too late. It's been noticed that customers who have exactly two banking products are more likely to stay with the bank. So, banks could try to sell other financial products to

customers who only have one product, but they need to be careful about which products they choose. On the other hand, customers who have many products might get confused, so banks should make it easier for them to manage everything and check in with them regularly to make sure they're doing okay. This way, banks can make things less complicated for their customers and show them that they care.

To keep customers who are middle-aged, banks should offer services that are just right for them. This could include helping them plan their money, get ready for retirement, and manage their wealth. Banks can also assign a special advisor to each customer, so they get personalized help and feel like the bank really cares about them. By doing this, banks can build strong relationships with these customers and keep them from going to another bank.

Because inactive customers are significantly more likely to churn, financial institutions should establish automated engagement systems that monitor transaction frequency and trigger personalized communications whenever customer activity declines. Promotional offers, loyalty rewards, and proactive relationship management may help re-engage these customers.

To really understand what's going on with customers in Germany, we need to take a closer look at how our service is doing in that area, how our prices compare to others, and where we stand against the competition. By doing some local marketing and checking in with customers to see how happy they are, we might be able to cut down on the number of people leaving us in that market. This could be a good way to make our business stronger and keep more customers.

11. Conclusion

This study created a new way to predict when customers might leave their bank, using a type of artificial intelligence called machine learning. The researchers used a large set of customer data - 10,000 records - and combined several techniques, including cleaning up the data, selecting the most important features, and using special algorithms to make predictions. They also used a method called SHAP to help explain why the model made certain predictions, which is important for understanding why customers might be at risk of leaving. By doing this, the study was able to identify which customers are most likely to switch to a different bank.

When we looked at the five different models we were testing, one really stood out - the Gradient Boosting classifier. It was the best at predicting what would happen, with an accuracy of 86.2% and a ROC-AUC score of 0.8652. But what's really interesting is what we found out about why customers stop doing business with us. It turns out that how old the customer is, how many banking products they use, whether they're an active member, where they live, and how they handle their money are all really important factors. These things have a big impact on whether a customer decides to leave or stay.

The new approach helps banks move away from just reacting to customer issues and instead use data to stay ahead. By finding out which customers are most likely to leave and explaining why, banks can use their marketing money more wisely, make customers happier, and ultimately make more money in the long run. This way, banks can focus on keeping the customers they already have, rather than just trying to get new ones. It's all about using information to make smarter decisions and build stronger relationships with customers.

To take this work to the next level, future studies could look into using real-time data on transactions, the history of how customers interact with the company, and advanced techniques like deep learning to make predictions even more accurate. Also, by putting live customer analytics dashboards and adaptive learning models into action, companies could keep a close eye on customers who might leave and make their customer relationship management systems more responsive. This way, they can stay on top of things and make changes as needed to keep their customers happy and engaged.